

Supplementary materials of “TabMaDo: Dual-Guided Tabulamba Diffusion Oversampling for Imbalanced Tabular Data Classification”

Part A: Proof of Convergence Analysis

.1 Assumptions of Convergence Analysis

Assumption 1 (Gradient Lipschitz Smoothness). *The gradients of L_{class} and L_{proto} are Lipschitz continuous. For any two parameter vectors θ_1, θ_2 , there exist positive constants $\beta_1 > 0, \beta_2 > 0$ such that:*

$$\|\nabla L_{class}(\theta_2) - \nabla L_{class}(\theta_1)\| \leq \beta_1 \|\theta_2 - \theta_1\|,$$

$$\|\nabla L_{proto}(\theta_2) - \nabla L_{proto}(\theta_1)\| \leq \beta_2 \|\theta_2 - \theta_1\|.$$

The composite loss L_G is Lipschitz smoothness with $\beta = \max(\beta_1, \gamma\beta_2)$.

Assumption 2 (Gradient Boundedness). *The expected squared norm of stochastic gradients for all parameters is uniformly bounded. Let $\theta \in \{\theta_x, \theta_{min}, \theta_h\}$ denote the parameter sets of E_x, E_{min} and C , respectively. There exists $G > 0$ such that:*

$$\mathbb{E} [\|\nabla_{\theta} L_{class}\|^2] \leq G, \quad \mathbb{E} [\|\nabla_{\theta} L_{proto}\|^2] \leq G.$$

Assumption 3 (Encoder Lipschitz Continuity). *Encoders E_x and E_{min} are L_E -Lipschitz continuous. For any inputs a, b :*

$$\|E_x(a) - E_x(b)\| \leq L_E \|a - b\|,$$

$$\|E_{min}(a) - E_{min}(b)\| \leq L_E \|a - b\|.$$

.2 Proof of Lemmas 1-3

Lemma 1 (Monotonicity of L_{class} with θ_h Update). *At iteration $t+1$, fix θ_x and θ_{min} , and only update θ_h . Then $L_{class}(t+1) \leq L_{class}(t)$, and L_{proto} remains unchanged.*

Proof. L_{class} is a binary cross-entropy loss with respect to θ_h , satisfying non-negativity: $L_{class} \geq 0$. By Assumption 2, its gradient norm is bounded: $\|\nabla_{\theta_h} L_{class}\| \leq \sqrt{G}$. The update rule for θ_h at iteration t is:

$$\theta_h(t+1) = \theta_h(t) - \eta(t) \cdot \nabla_{\theta_h} L_{class}(\theta_h(t)),$$

where $\eta(t) > 0$ denotes the learning rate at iteration t . Under Assumption 1 (Lipschitz smoothness of L_{class}), for any two parameter vectors $\theta_h(t)$ and $\theta_h(t+1)$, the following inequality holds:

$$\begin{aligned} L_{class}(\theta_h(t+1)) &\leq L_{class}(\theta_h(t)) \\ &\quad + \langle \nabla L_{class}(\theta_h(t)), \Delta\theta_h \rangle + \frac{\beta_1}{2} \|\Delta\theta_h\|^2, \end{aligned}$$

where $\Delta\theta_h = \theta_h(t+1) - \theta_h(t) = -\eta(t)\nabla_{\theta_h} L_{class}(\theta_h(t))$. Substitute $\Delta\theta_h$ into the above inequality:

$$\begin{aligned} L_{class}(t+1) &\leq L_{class}(t) - \eta(t) \|\nabla_{\theta_h} L_{class}\|^2 \\ &\quad + \frac{\beta_1 \eta(t)^2}{2} \|\nabla_{\theta_h} L_{class}\|^2. \end{aligned}$$

Set the learning rate constraint $\eta(t) \leq \frac{1}{\beta_1}$, and the inequality can be further simplified as:

$$L_{class}(t+1) \leq L_{class}(t) - \frac{\eta(t)}{2} \|\nabla_{\theta_h} L_{class}\|^2 \leq L_{class}(t).$$

Since L_{proto} is independent of θ_h , it remains unchanged:

$$L_{proto}(t+1) = L_{proto}(t).$$

Thus, the overall loss satisfies:

$$L_G(t+1) = L_{class}(t+1) + \gamma L_{proto}(t+1) < L_G(t).$$

Hence, Lemma 1 is proved. $\square$

Lemma 2 (Monotonicity of L_{proto} with θ_{min} Update). *At iteration $t+1$, fix θ_x and θ_h , and only update the prototype encoder parameter θ_{min} . Then $L_{proto}(t+1) \leq L_{proto}(t)$, and L_{class} remains unchanged.*

Proof. The prototype loss L_{proto} depends on the cosine similarity $sim = \cos(h_x, h_{min})$, where $h_{min} = E_{min}(p_{min})$. Updating θ_{min} modifies h_{min} , thereby affecting the value of L_{proto} . By Assumption 2, the gradient norm of L_{proto} with respect to θ_{min} satisfies $\|\nabla_{\theta_{min}} L_{proto}\| \leq \sqrt{G}$. The gradient descent update rule for θ_{min} is:

$$\theta_{min}(t+1) = \theta_{min}(t) - \eta(t) \nabla_{\theta_{min}} L_{proto}(\theta_{min}(t)).$$

Note that L_{proto} is a sum of two convex-like terms: (i) $1 - sim$ for minority samples, which is convex in sim ; (ii) $ReLU(sim)$ for majority samples, which is convex and non-decreasing. Since $sim = \cos(h_x, h_{min})$ is Lipschitz in h_{min} (by Assumption 3), and the overall composition preserves Lipschitz smoothness, the following descent property holds with learning rate constraint $\eta(t) \leq \frac{1}{\beta_2}$:

$$L_{proto}(t+1) \leq L_{proto}(t) - \frac{\eta(t)}{2} \|\nabla_{\theta_{min}} L_{proto}\|^2 \leq L_{proto}(t).$$

Since L_{class} is independent of θ_{min} , it remains unchanged. Thus, the composite loss satisfies:

$$L_G(t+1) = L_{class}(t+1) + \gamma L_{proto}(t+1) < L_G(t).$$

Hence, Lemma 2 is proved. $\square$

Lemma 3 (Monotonicity of Composite Loss with θ_x Update). *At iteration $t+1$, fix θ_h and θ_{min} , and update the sample encoder parameter θ_x . The total gradient of the composite loss with respect to θ_x is $\nabla_{\theta_x} L_G = \nabla_{\theta_x} L_{class} + \gamma \nabla_{\theta_x} L_{proto}$, and the composite loss satisfies $L_G(t+1) \leq L_G(t)$.*

Proof. By the Cauchy-Schwarz inequality and Assumption 2, the squared norm of the total gradient can be bounded as:

$$\begin{aligned} \|\nabla_{\theta_x} L_G\|^2 &= \|\nabla_{\theta_x} L_{class} + \gamma \nabla_{\theta_x} L_{proto}\|^2 \\ &\leq 2 \left(\|\nabla_{\theta_x} L_{class}\|^2 + \gamma^2 \|\nabla_{\theta_x} L_{proto}\|^2 \right) \\ &\leq 2(1 + \gamma^2)G. \end{aligned}$$

Performing gradient descent with step size $\eta(t)$:

$$\theta_x(t+1) = \theta_x(t) - \eta(t) \cdot \nabla_{\theta_x} L_G(\theta_x(t)).$$

Based on Assumption 1 (Lipschitz smoothness of L_{class} and L_{proto}), with the learning rate constraint $\eta(t) \leq \frac{1}{\beta}$, the following inequalities for the two loss components can be derived:

$$L_{class}(t+1) \leq L_{class}(t) - \frac{\eta(t)}{2} \|\nabla_{\theta_x} L_{class}\|^2,$$

$$L_{proto}(t+1) \leq L_{proto}(t) - \frac{\eta(t)}{2} \|\nabla_{\theta_x} L_{proto}\|^2.$$

Multiply the second inequality by γ and sum the two inequalities:

$$\begin{aligned} L_G(t+1) &\leq L_G(t) - \frac{\eta(t)}{2} \left(\|\nabla_{\theta_x} L_{class}\|^2 + \gamma^2 \|\nabla_{\theta_z} L_{proto}\|^2 \right) \\ &\leq L_G(t). \end{aligned}$$

Lemma 3 is proved. $\square$

.3 Proof of Theorem 1

Theorem 1 (Convergence of the Overall Loss). *Under Assumptions 1–3, set the learning rate $\eta = \frac{1}{\beta_\xi}$ where $\xi = 2(1 + \gamma^2)G$. For the composite loss sequence $\{L_G(\theta_t)\}$ generated by gradient descent updates, the loss sequence converges to a constant $Loss^*$, i.e., $\lim_{t \rightarrow \infty} L_G(\theta_t) = Loss^*$. Meanwhile, the gradient norm of the loss function converges to 0: $\lim_{t \rightarrow \infty} \|\nabla L_G(\theta_t)\| = 0$.*

Proof. Lemmas 1–3 prove that the composite loss is non-increasing under updates of any parameter component. Since L_{class} and L_{proto} are non-negative cross-entropy losses, $L_G(\theta_t) \geq 0$ for all t . By the Monotone Convergence Theorem, the bounded non-increasing sequence $\{L_G(\theta_t)\}$ converges to a limit $Loss^*$. From the descent inequality $L_G(\theta_t) - L_G(\theta_{t+1}) \geq \frac{\eta(t)}{2} \|\nabla L_G(\theta_t)\|^2$, summing from $t = 0$ to $t = T - 1$ gives:

$$L_G(\theta_0) - L_G(\theta_T) \geq \frac{\eta}{2} \sum_{t=0}^{T-1} \|\nabla L_G(\theta_t)\|^2.$$

Since the left-hand side is bounded by $L_G(\theta_0) - Loss^*$, the series $\sum_{t=0}^{\infty} \|\nabla L_G(\theta_t)\|^2$ converges, implying $\lim_{t \rightarrow \infty} \|\nabla L_G(\theta_t)\| = 0$. $\square$

Part B: Additional Experiment Results

.1 Performance of Additional Classifiers

This part provides detailed performance metrics for GBDT and TabPFN, provided in Tables 1 and 2. Each table reports the mean and standard deviation of the corresponding metric across the 14 benchmark datasets. The best result for each dataset is highlighted in bold with a green background.

.2 Ablation Study Results on Additional Classifiers

This part provides the detailed results of the ablation study for the two classifiers GBDT and TabPFN. Tables 3 and 4, report the mean F1-score, G-mean, and AUC of the full TabMaDo framework compared to its four ablated variants across twelve datasets. The best result for each dataset and metric is highlighted in bold with a green background.

Table 1: Performance Comparison of Seven Oversampling Methods on GBDT.

Metric	Dataset	ssmote	tabddpm	codi	tabdit	relddpm	GOIO	TabMaDo
F1-score	yeast6	0.4402±0.1687	0.3829±0.1975	0.3317±0.2734	0.3783±0.3259	0.5548±0.1502	0.4187±0.2128	0.5743±0.2038
	flare-F	0.2586±0.0976	0.2225±0.1262	0.1402±0.1384	0.2891±0.2017	0.3440±0.1307	0.3042±0.1127	0.3197±0.2353
	wilt	0.8498±0.0416	0.8131±0.0375	0.7756±0.0585	0.8416±0.0304	0.8654±0.0295	0.1533±0.0591	0.8658±0.0312
	abalone9-18	0.3794±0.1433	0.4659±0.1925	0.3688±0.1859	0.3276±0.2008	0.4285±0.1818	0.3023±0.1808	0.5002±0.1950
	ecoli4	0.7733±0.2878	0.7433±0.2989	0.7033±0.3150	0.7800±0.2982	0.7933±0.2958	0.5267±0.2984	0.7433±0.2989
	glass-0-1-4-6-vs-2	0.3271±0.2840	0.3467±0.2971	0.2500±0.3096	0.2167±0.3500	0.3619±0.2809	0.0500±0.1500	0.3700±0.2934
	led7digit	0.6677±0.1812	0.6170±0.1951	0.3953±0.0656	0.3896±0.0736	0.7068±0.1840	0.0750±0.1601	0.5940±0.2480
	vehicle0	0.9464±0.0260	0.9424±0.0324	0.9272±0.0342	0.9351±0.0283	0.9429±0.0375	0.6101±0.1752	0.9400±0.0308
	haberman	0.4448±0.1123	0.3378±0.0895	0.3563±0.1537	0.3010±0.1003	0.3906±0.0857	0.3214±0.1200	0.4465±0.0790
	phoneme	0.7629±0.0241	0.7661±0.0217	0.7552±0.0203	0.7685±0.0241	0.7714±0.0108	0.2356±0.0438	0.7716±0.0170
	german	0.5581±0.0626	0.5490±0.0574	0.5225±0.0596	0.5328±0.0454	0.5586±0.0569	0.5732±0.0693	0.5598±0.0838
	saheart	0.4934±0.0766	0.4990±0.0571	0.5076±0.1089	0.5074±0.0865	0.5316±0.0704	0.5504±0.0677	0.5508±0.0824
	pima	0.6546±0.0866	0.6491±0.0743	0.6194±0.0786	0.6197±0.0704	0.6502±0.0873	0.5292±0.0882	0.6640±0.0560
	magic	0.8156±0.0077	0.8163±0.0081	0.8022±0.0090	0.8020±0.0080	0.8162±0.0067	0.6148±0.2089	0.8176±0.0086
G-mean	yeast6	0.7658±0.1423	0.6670±0.1877	0.4553±0.3262	0.4817±0.3515	0.7768±0.1373	0.7617±0.1489	0.7850±0.1558
	flare-F	0.7035±0.1558	0.5949±0.2415	0.3017±0.2560	0.4603±0.2553	0.6798±0.1417	0.7182±0.1599	0.5609±0.3167
	wilt	0.9609±0.0225	0.9607±0.0216	0.8398±0.0543	0.8958±0.0283	0.9547±0.0310	0.4541±0.0837	0.9658±0.0196
	abalone9-18	0.6446±0.1074	0.6294±0.1551	0.4787±0.1837	0.4289±0.2322	0.6602±0.1528	0.5904±0.2440	0.7288±0.1342
	ecoli4	0.8627±0.2998	0.8078±0.2992	0.8300±0.3009	0.8366±0.3010	0.8642±0.3005	0.6980±0.3708	0.8078±0.2992
	glass-0-1-4-6-vs-2	0.4624±0.3897	0.4670±0.3941	0.2809±0.3440	0.2395±0.3742	0.5069±0.3419	0.0687±0.2062	0.5067±0.3450
	led7digit	0.8721±0.1263	0.8472±0.1167	0.8329±0.0592	0.8058±0.0731	0.8775±0.1251	0.1059±0.2128	0.7737±0.2900
	vehicle0	0.9722±0.0184	0.9646±0.0232	0.9543±0.0286	0.9604±0.0221	0.9680±0.0295	0.6723±0.1402	0.9735±0.0214
	haberman	0.5917±0.0995	0.5012±0.0830	0.4887±0.1828	0.4590±0.0959	0.5464±0.0838	0.4829±0.1115	0.5929±0.0702
	phoneme	0.8532±0.0172	0.8510±0.0151	0.8211±0.0156	0.8334±0.0201	0.8521±0.0080	0.3790±0.0391	0.8550±0.0131
	german	0.7079±0.0505	0.6577±0.0454	0.6378±0.0498	0.6426±0.0336	0.6728±0.0469	0.6870±0.0594	0.6774±0.0686
	saheart	0.5922±0.0658	0.6015±0.0466	0.6099±0.0895	0.6109±0.0708	0.6295±0.0573	0.6392±0.0595	0.6439±0.0686
	pima	0.7302±0.0720	0.7241±0.0606	0.6998±0.0624	0.7000±0.0552	0.7266±0.0714	0.6261±0.0713	0.7380±0.0455
	magic	0.8522±0.0055	0.8522±0.0061	0.8355±0.0066	0.8351±0.0062	0.8520±0.0055	0.6637±0.2238	0.8523±0.0069
AUC	yeast6	0.9176±0.0501	0.8791±0.0741	0.8993±0.0691	0.8936±0.0686	0.9271±0.0531	0.8991±0.0798	0.9354±0.0592
	flare-F	0.8219±0.1250	0.8311±0.1183	0.8533±0.0709	0.8565±0.0935	0.8547±0.1144	0.8615±0.0978	0.8914±0.0847
	wilt	0.9933±0.0081	0.9915±0.0127	0.9886±0.0067	0.9908±0.0114	0.9946±0.0066	0.7426±0.0325	0.9957±0.0048
	abalone9-18	0.8334±0.0730	0.8437±0.1084	0.8047±0.1151	0.8188±0.0914	0.8438±0.0751	0.7806±0.1414	0.8514±0.0840
	ecoli4	0.9452±0.1592	0.9920±0.0194	0.9921±0.0128	0.9841±0.0296	0.9871±0.0337	0.8991±0.1261	0.9839±0.0382
	glass-0-1-4-6-vs-2	0.8941±0.0654	0.8911±0.1087	0.6601±0.2533	0.7956±0.1761	0.9175±0.0938	0.5854±0.1743	0.8984±0.0756
	led7digit	0.9633±0.0333	0.9581±0.0347	0.9054±0.0527	0.8948±0.0743	0.9659±0.0328	0.2740±0.1142	0.8930±0.1383
	vehicle0	0.9960±0.0044	0.9962±0.0036	0.9948±0.0040	0.9945±0.0048	0.9943±0.0069	0.9122±0.0496	0.9956±0.0040
	haberman	0.6267±0.1262	0.6207±0.0683	0.6339±0.1042	0.6247±0.0833	0.6351±0.0871	0.5837±0.0671	0.6297±0.0851
	phoneme	0.9226±0.0099	0.9249±0.0118	0.9213±0.0086	0.9257±0.0102	0.9271±0.0088	0.7131±0.0385	0.9264±0.0102
	german	0.7810±0.0451	0.7800±0.0470	0.7775±0.0412	0.7799±0.0450	0.7771±0.0444	0.7776±0.0703	0.7628±0.0706
	saheart	0.6662±0.0648	0.6989±0.0440	0.7119±0.0536	0.7092±0.0585	0.7098±0.0510	0.7066±0.0580	0.7020±0.0454
	pima	0.8111±0.0535	0.8106±0.0448	0.8012±0.0457	0.8074±0.0397	0.8128±0.0421	0.7473±0.0598	0.8168±0.0447
	magic	0.9301±0.0059	0.9295±0.0055	0.9263±0.0062	0.9273±0.0061	0.9296±0.0059	0.7764±0.2592	0.9291±0.0061

Table 2: Performance Comparison of Seven Oversampling Methods on TabPFN.

Metric	Dataset	ssmote	tabddpm	codi	tabdit	relddp	GOIO	TabMaDo
F1-score	yeast6	0.1400±0.2279	0.3973±0.2153	0.4857±0.2011	0.3686±0.2964	0.5342±0.1954	0.3389±0.2681	0.5349±0.1284
	flare-F	0.2133±0.1231	0.2305±0.1513	0.1367±0.1600	0.2372±0.2765	0.3311±0.1130	0.3263±0.1382	0.3919±0.2015
	wilt	0.9872±0.0170	0.9825±0.0221	0.9924±0.0116	0.9797±0.0248	0.9828±0.0221	0.9499±0.0457	0.9927±0.0220
	abalone9-18	0.0800±0.1600	0.5392±0.1114	0.5394±0.2359	0.5037±0.1270	0.5585±0.1417	0.3862±0.2163	0.5479±0.1030
	ecoli4	0.5467±0.3781	0.8123±0.3023	0.8133±0.3023	0.8167±0.3027	0.7567±0.2982	0.5471±0.3167	0.8173±0.3023
	glass-0-1-4-6-vs-2	0.0500±0.1500	0.1833±0.3202	0.1567±0.2468	0.3167±0.3202	0.0667±0.2000	0.0400±0.1200	0.3633±0.3437
	led7digit	0.6950±0.1777	0.6597±0.2012	0.3837±0.0563	0.3807±0.0859	0.7673±0.1846	0.0500±0.1500	0.6124±0.2700
	vehicle0	0.9872±0.0170	0.9825±0.0221	0.9924±0.0116	0.9797±0.0248	0.9828±0.0221	0.9499±0.0457	0.9927±0.0220
	haberman	0.5007±0.1427	0.4721±0.0720	0.3052±0.1946	0.2952±0.1476	0.4887±0.1522	0.2774±0.1107	0.5356±0.1072
	phoneme	0.7843±0.2618	0.8730±0.0127	0.8782±0.0162	0.8799±0.0126	0.8709±0.0117	0.6744±0.0315	0.8725±0.0138
	german	0.4888±0.1655	0.5722±0.0688	0.5406±0.0737	0.5729±0.0600	0.5823±0.0589	0.5737±0.0733	0.5838±0.0631
	saheart	0.5275±0.0489	0.5865±0.0770	0.5666±0.1099	0.5386±0.1017	0.5338±0.1078	0.5870±0.0752	0.5981±0.0641
	pima	0.6437±0.0784	0.6645±0.0783	0.6207±0.0834	0.6094±0.0772	0.6346±0.0732	0.5147±0.0661	0.6606±0.0533
	magic	0.8405±0.0083	0.8578±0.0067	0.8549±0.0055	0.8553±0.0052	0.8553±0.0048	0.7849±0.0318	0.8560±0.0054
G-mean	yeast6	0.1705±0.2660	0.5866±0.2255	0.6239±0.1593	0.4410±0.3105	0.7983±0.1546	0.6307±0.2658	0.8362±0.1268
	flare-F	0.5608±0.2965	0.5207±0.2965	0.2558±0.2649	0.3200±0.3377	0.6399±0.1006	0.6431±0.1781	0.6576±0.2681
	wilt	0.9890±0.0136	0.9892±0.0139	0.9958±0.0081	0.9865±0.0187	0.9911±0.0129	0.9614±0.0399	0.9960±0.0121
	abalone9-18	0.1000±0.2000	0.6882±0.1064	0.6452±0.2580	0.6218±0.1171	0.7582±0.1284	0.6163±0.2382	0.8169±0.0812
	ecoli4	0.5813±0.3988	0.8398±0.3025	0.8398±0.3025	0.8591±0.3024	0.8355±0.3014	0.6991±0.3709	0.8659±0.3012
	glass-0-1-4-6-vs-2	0.0688±0.2065	0.2336±0.3671	0.2064±0.3154	0.3517±0.3517	0.0707±0.2121	0.0669±0.2007	0.4686±0.3990
	led7digit	0.8768±0.1270	0.8662±0.1228	0.8274±0.0534	0.8000±0.0820	0.8857±0.1273	0.0577±0.1732	0.7761±0.2915
	vehicle0	0.9890±0.0136	0.9892±0.0139	0.9958±0.0081	0.9865±0.0187	0.9911±0.0129	0.9614±0.0399	0.9960±0.0121
	haberman	0.6470±0.1232	0.6176±0.0605	0.4334±0.2009	0.4186±0.1717	0.6324±0.1290	0.4360±0.0963	0.6745±0.0933
	phoneme	0.8140±0.2716	0.9181±0.0072	0.9101±0.0113	0.9122±0.0106	0.9160±0.0090	0.7300±0.0243	0.9201±0.0112
	german	0.5831±0.1960	0.6714±0.0547	0.6454±0.0594	0.6704±0.0491	0.6912±0.0473	0.6821±0.0644	0.6958±0.0513
	saheart	0.6237±0.0396	0.6749±0.0631	0.6568±0.0898	0.6336±0.0831	0.6322±0.0882	0.6627±0.0682	0.6825±0.0540
	pima	0.7209±0.0658	0.7326±0.0651	0.6984±0.0681	0.6884±0.0636	0.7127±0.0609	0.6129±0.0528	0.7340±0.0443
	magic	0.8733±0.0060	0.8829±0.0051	0.8784±0.0041	0.8786±0.0044	0.8815±0.0038	0.8176±0.0275	0.8830±0.0043
AUC	yeast6	0.4051±0.4138	0.7839±0.2806	0.9284±0.0597	0.9129±0.0656	0.9343±0.0611	0.8958±0.0810	0.9404±0.0665
	flare-F	0.8553±0.0749	0.8549±0.0650	0.8635±0.0603	0.8868±0.0705	0.8573±0.1012	0.8947±0.0668	0.8952±0.0755
	wilt	0.9997±0.0007	0.9996±0.0006	0.9997±0.0005	0.9995±0.0011	0.9995±0.0010	0.9964±0.0057	0.9997±0.0009
	abalone9-18	0.8934±0.1049	0.9227±0.0559	0.9461±0.0548	0.9336±0.0567	0.9362±0.0487	0.8792±0.0773	0.9477±0.0365
	ecoli4	0.9694±0.0867	0.9912±0.0103	0.9920±0.0194	0.9932±0.0148	0.9920±0.0194	0.9607±0.0511	0.9936±0.0148
	glass-0-1-4-6-vs-2	0.8841±0.1077	0.9278±0.0421	0.8620±0.1408	0.9072±0.0744	0.8692±0.1211	0.7101±0.1133	0.9098±0.0873
	led7digit	0.9491±0.0471	0.9515±0.0477	0.9080±0.0522	0.9068±0.0461	0.9602±0.0361	0.2532±0.1412	0.8773±0.1755
	vehicle0	0.9997±0.0007	0.9996±0.0006	0.9997±0.0005	0.9995±0.0011	0.9995±0.0010	0.9964±0.0057	0.9997±0.0009
	haberman	0.6840±0.1203	0.6900±0.0953	0.5914±0.2387	0.6797±0.0960	0.7042±0.1327	0.5987±0.0814	0.7266±0.0993
	phoneme	0.8794±0.2932	0.9766±0.0036	0.9766±0.0039	0.9764±0.0040	0.9768±0.0045	0.9258±0.0152	0.9774±0.0038
	german	0.7066±0.2403	0.8046±0.0586	0.8060±0.0574	0.8082±0.0630	0.7833±0.0512	0.7882±0.0613	0.7878±0.0586
	saheart	0.7061±0.0687	0.7616±0.0562	0.7612±0.0590	0.7564±0.0628	0.7417±0.0554	0.7467±0.0574	0.7624±0.0500
	pima	0.8210±0.0406	0.8339±0.0471	0.8361±0.0382	0.8360±0.0351	0.8313±0.0391	0.7587±0.0545	0.8398±0.0340
	magic	0.9446±0.0034	0.9540±0.0042	0.9542±0.0040	0.9541±0.0038	0.9536±0.0037	0.9239±0.0125	0.9535±0.0040

Table 3: Ablation Study Results of TabMaDo Across Fourteen Benchmark Datasets on GBDT.

Metric	Dataset	V1	V2	V3	V4	TabMaDo
F1-score	yeast6	0.4383±0.2242	0.3550±0.2642	0.3467±0.2778	0.3355±0.2702	0.5743±0.2038
	wilt	0.8627±0.0321	0.8572±0.0320	0.8442±0.0488	0.8580±0.0386	0.8658±0.0312
	abalone9-18	0.4162±0.1495	0.3565±0.1743	0.3762±0.1003	0.4002±0.1827	0.5002±0.1950
	ecoli4	0.7100±0.2868	0.7133±0.2782	0.7767±0.3070	0.7933±0.2958	0.7433±0.2989
	glass-0-1-4-6-vs-2	0.3533±0.3067	0.1167±0.2363	0.2133±0.3277	0.1300±0.2685	0.3700±0.2934
	vehicle0	0.9394±0.0161	0.9488±0.0374	0.9370±0.0454	0.9315±0.0394	0.9400±0.0308
	haberman	0.4674±0.0868	0.2956±0.1194	0.3057±0.1401	0.3119±0.1646	0.4465±0.0790
	phoneme	0.7780±0.0153	0.7641±0.0229	0.7541±0.0316	0.7702±0.0260	0.7716±0.0170
	german	0.5835±0.0685	0.5222±0.0940	0.5171±0.0509	0.5365±0.0709	0.5598±0.0838
	saheart	0.5286±0.0607	0.4741±0.1061	0.4615±0.1129	0.4708±0.1059	0.5508±0.0824
	pima	0.6446±0.0704	0.6153±0.0771	0.6126±0.0783	0.6238±0.0647	0.6640±0.0560
	magic	0.8194±0.0087	0.8052±0.0084	0.8061±0.0084	0.8080±0.0096	0.8176±0.0086
G-mean	yeast6	0.6286±0.2558	0.4683±0.3330	0.4405±0.3101	0.4477±0.3121	0.7850±0.1558
	wilt	0.9602±0.0242	0.9103±0.0346	0.9015±0.0469	0.9082±0.0400	0.9658±0.0196
	abalone9-18	0.6690±0.1443	0.4779±0.1833	0.5441±0.0893	0.5595±0.1257	0.7288±0.1342
	ecoli4	0.8045±0.2972	0.7519±0.2853	0.8371±0.3022	0.8382±0.3017	0.8078±0.2992
	glass-0-1-4-6-vs-2	0.4697±0.3975	0.1394±0.2789	0.2388±0.3711	0.1662±0.3384	0.5067±0.3450
	vehicle0	0.9718±0.0153	0.9645±0.0331	0.9609±0.0337	0.9541±0.0309	0.9735±0.0214
	haberman	0.6107±0.0848	0.4480±0.1081	0.4606±0.1309	0.4454±0.1888	0.5929±0.0702
	phoneme	0.8581±0.0102	0.8255±0.0210	0.8182±0.0267	0.8292±0.0213	0.8550±0.0131
	german	0.6936±0.0554	0.6273±0.0754	0.6270±0.0409	0.6435±0.0583	0.6774±0.0686
	saheart	0.6262±0.0478	0.5841±0.0869	0.5742±0.0920	0.5802±0.0883	0.6439±0.0686
	pima	0.7228±0.0568	0.6966±0.0610	0.6944±0.0632	0.7032±0.0526	0.7380±0.0455
	magic	0.8547±0.0067	0.8366±0.0067	0.8374±0.0070	0.8391±0.0077	0.8523±0.0069
AUC	yeast6	0.9290±0.0547	0.9107±0.0568	0.9166±0.0580	0.9045±0.0681	0.9354±0.0592
	wilt	0.9935±0.0098	0.9939±0.0038	0.9934±0.0040	0.9867±0.0158	0.9957±0.0048
	abalone9-18	0.8450±0.0946	0.8103±0.1305	0.8502±0.0826	0.8169±0.0811	0.8514±0.0840
	ecoli4	0.9936±0.0127	0.9872±0.0295	0.9792±0.0479	0.9808±0.0393	0.9839±0.0382
	glass-0-1-4-6-vs-2	0.8819±0.1031	0.8484±0.1564	0.9038±0.0487	0.8671±0.0779	0.8984±0.0756
	vehicle0	0.9946±0.0052	0.9955±0.0048	0.9957±0.0041	0.9948±0.0045	0.9956±0.0040
	haberman	0.6584±0.0860	0.6188±0.0805	0.6319±0.0951	0.6120±0.0933	0.6297±0.0851
	phoneme	0.9274±0.0085	0.9305±0.0089	0.9279±0.0093	0.9328±0.0084	0.9264±0.0102
	german	0.7819±0.0598	0.7860±0.0585	0.7916±0.0521	0.7846±0.0546	0.7628±0.0706
	saheart	0.7033±0.0631	0.6815±0.0753	0.7021±0.0651	0.6930±0.0599	0.7020±0.0454
	pima	0.8164±0.0402	0.8124±0.0484	0.8077±0.0504	0.8110±0.0455	0.8168±0.0447
	magic	0.9298±0.0057	0.9280±0.0057	0.9278±0.0059	0.9300±0.0051	0.9291±0.0061

Table 4: Ablation Study Results of TabMaDo Across Fourteen Benchmark Datasets on TabPFN

Metric	Dataset	V1	V2	V3	V4	TabMaDo
F1-score	yeast6	0.5277±0.1866	0.4671±0.2178	0.4533±0.1249	0.4433±0.3019	0.5349±0.1284
	wilt	0.8699±0.0355	0.9021±0.0255	0.9059±0.0326	0.9074±0.0318	0.9927±0.0220
	abalone9-18	0.4332±0.1568	0.5432±0.2211	0.6071±0.1521	0.5062±0.1690	0.5479±0.1030
	ecoli4	0.8133±0.3023	0.7800±0.2982	0.8133±0.3023	0.8133±0.3023	0.8173±0.3023
	glass-0-1-4-6-vs-2	0.1733±0.2736	0.2067±0.2603	0.1567±0.2468	0.1467±0.2948	0.3633±0.3437
	vehicle0	0.9850±0.0197	0.9800±0.0241	0.9850±0.0252	0.9806±0.0300	0.9927±0.0220
	haberman	0.5011±0.1347	0.2998±0.1763	0.2543±0.1425	0.3955±0.1606	0.5356±0.1072
	phoneme	0.8802±0.0157	0.8728±0.0094	0.8638±0.0103	0.8753±0.0145	0.8725±0.0138
	german	0.5891±0.0620	0.5138±0.0642	0.5110±0.0596	0.5409±0.0880	0.5838±0.0631
	saheart	0.4762±0.1068	0.5335±0.1035	0.5372±0.0876	0.5203±0.0885	0.5981±0.0641
	pima	0.6599±0.0637	0.6347±0.0816	0.6207±0.0800	0.6434±0.0681	0.6606±0.0533
	magic	0.8553±0.0062	0.8481±0.0069	0.8461±0.0061	0.8521±0.0066	0.8560±0.0054
G-mean	yeast6	0.7370±0.1608	0.5490±0.2191	0.5620±0.0922	0.5173±0.3003	0.8362±0.1268
	wilt	0.9698±0.0229	0.9479±0.0279	0.9574±0.0265	0.9516±0.0348	0.9960±0.0121
	abalone9-18	0.6782±0.1422	0.6345±0.2483	0.7175±0.1505	0.6344±0.1477	0.8169±0.0812
	ecoli4	0.8659±0.3012	0.8105±0.2998	0.8398±0.3025	0.8398±0.3025	0.8659±0.3012
	glass-0-1-4-6-vs-2	0.2083±0.3183	0.2751±0.3371	0.2064±0.3154	0.1680±0.3413	0.4686±0.3990
	vehicle0	0.9900±0.0129	0.9867±0.0154	0.9900±0.0156	0.9904±0.0161	0.9960±0.0121
	haberman	0.6401±0.1130	0.4206±0.1868	0.3816±0.1551	0.5261±0.1342	0.6745±0.0933
	phoneme	0.9220±0.0107	0.9031±0.0098	0.8979±0.0094	0.9062±0.0124	0.9201±0.0112
	german	0.6966±0.0508	0.6191±0.0518	0.6171±0.0502	0.6405±0.0712	0.6958±0.0513
	saheart	0.5856±0.0871	0.6301±0.0856	0.6313±0.0729	0.6202±0.0724	0.6825±0.0540
	pima	0.7326±0.0520	0.7095±0.0669	0.6978±0.0656	0.7153±0.0541	0.7340±0.0443
	magic	0.8816±0.0048	0.8716±0.0059	0.8702±0.0049	0.8749±0.0052	0.8830±0.0043
AUC	yeast6	0.9241±0.0722	0.9396±0.0575	0.9311±0.0634	0.9352±0.0594	0.9404±0.0665
	wilt	0.9944±0.0097	0.9962±0.0050	0.9943±0.0108	0.9953±0.0080	0.9997±0.0009
	abalone9-18	0.9164±0.0535	0.9462±0.0534	0.9510±0.0598	0.9508±0.0404	0.9477±0.0365
	ecoli4	0.9920±0.0194	0.9904±0.0241	0.9872±0.0247	0.9888±0.0289	0.9936±0.0148
	glass-0-1-4-6-vs-2	0.8791±0.1334	0.9020±0.0821	0.8830±0.0773	0.9016±0.0634	0.9098±0.0873
	vehicle0	0.9996±0.0007	0.9995±0.0010	0.9996±0.0006	0.9996±0.0009	0.9997±0.0009
	haberman	0.7020±0.1190	0.7211±0.0773	0.7213±0.0936	0.7101±0.1115	0.7266±0.0993
	phoneme	0.9773±0.0044	0.9771±0.0029	0.9763±0.0032	0.9770±0.0041	0.9774±0.0038
	german	0.7851±0.0491	0.7993±0.0527	0.7914±0.0572	0.8015±0.0521	0.7878±0.0586
	saheart	0.7301±0.0650	0.7609±0.0512	0.7629±0.0607	0.7362±0.0616	0.7624±0.0500
	pima	0.8342±0.0364	0.8412±0.0356	0.8416±0.0350	0.8397±0.0325	0.8398±0.0340
	magic	0.9539±0.0039	0.9512±0.0043	0.9511±0.0040	0.9526±0.0043	0.9535±0.0040